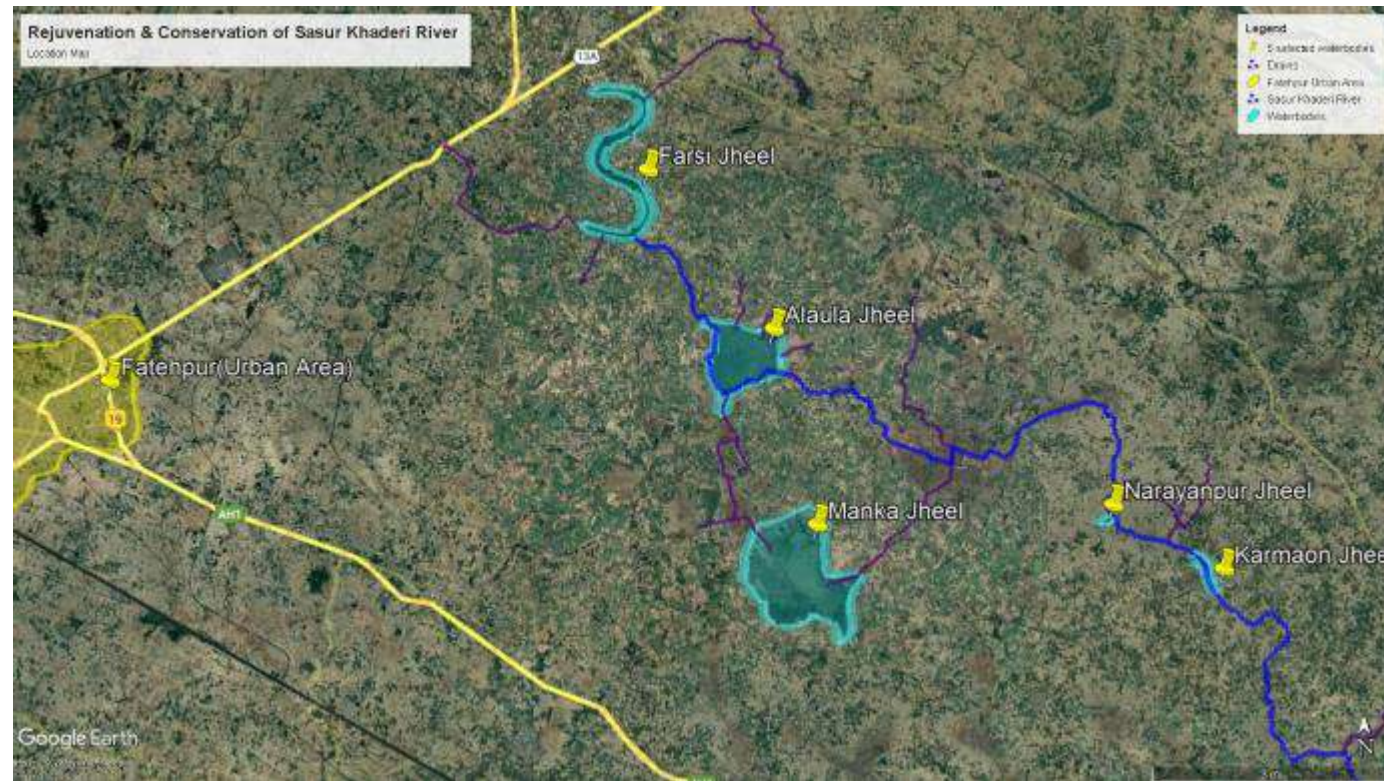


SURVEY, INVESTGATION & DPR PREPARATION FOR REJUVENATION OF WHOLE STRETCH OF SASUR KHADERI RIVER AND CONSERVATION OF FARSI AKHNAI JHEEL IN DISTRICT- FATEHPUR, UTTAR PRADESH



National Mission for Clean Ganga

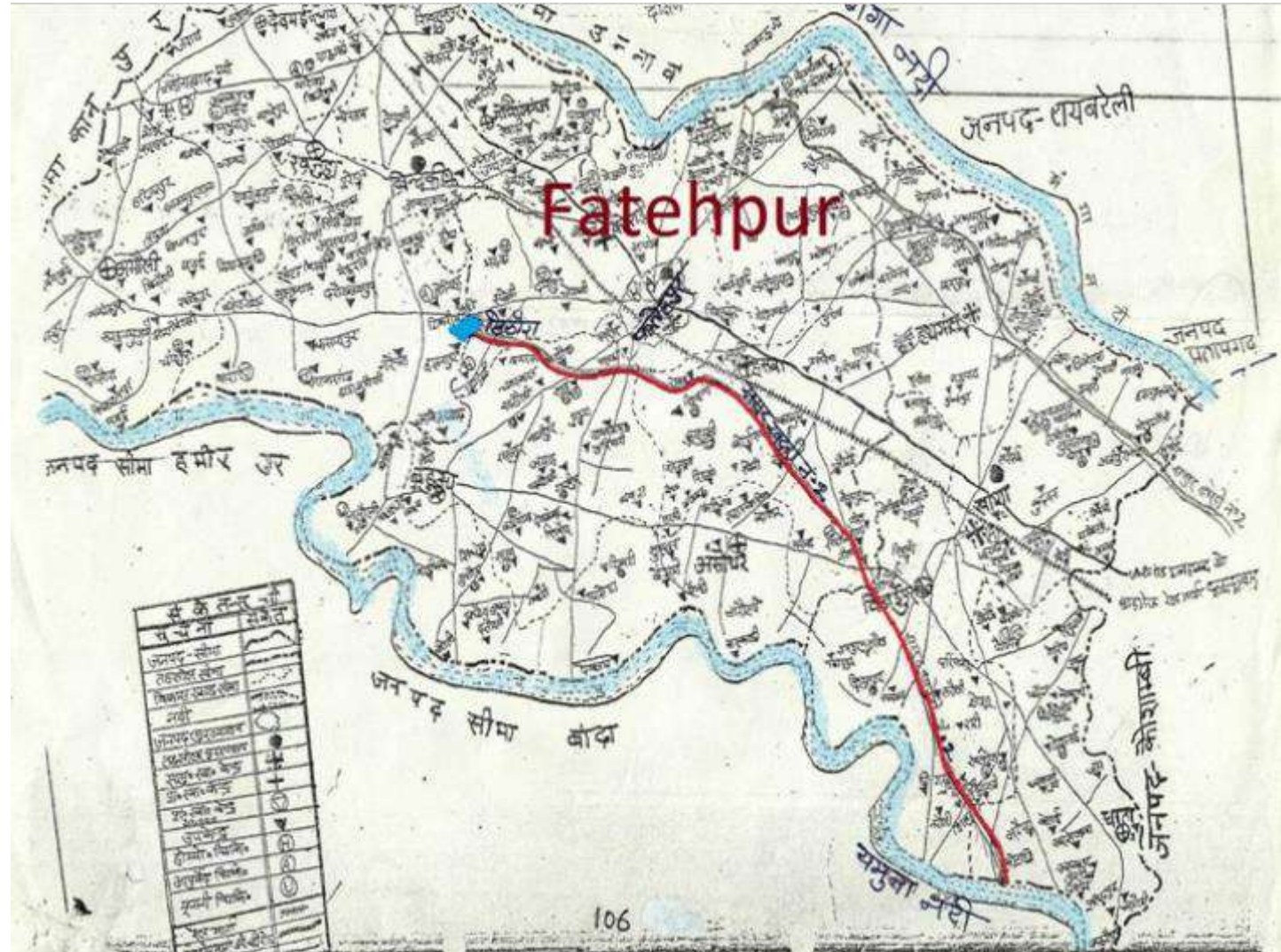
About Sasur Khederi river

- Tributary of the Yamuna River in UP and joins the Yamuna river 04 km before its confluence with the Ganga river at the Saraswati ghat in village Virampur in Prayagraj
- Flows through three districts of Uttar Pradesh i.e., Fatehpur, Kaushambhi and Prayagraj
- River originates from a series of waterbodies (lakes and ponds) in the Bhithaura block, Fatehpur Dist.
- River is being ***encroached to extinction***. For the past many years, the village panchayats through which the river passes have been dumping mud into it in an attempt to landfill. The reclaimed land is then sold to interested parties by issuing illegal documents of ownership. These parties end up using the land to build their houses or agricultural land.



Why Sasur Khaderi River ?

- Tributary to Yamuna and contribute flow to Ganga- Aviral Dhara
- The Irrigation Department, Govt. of UP, has submitted revised DPR (revised according to NMCG mandate, suggestions & recommendations) for rejuvenation of 5 ponds/ Jheels in Upstream dated 18th July 2022 which shall contribute to restoration of river.
- Historical and religious importance, as confirmed by locals.
- Fall under Small river rejuvenation initiative of NMCG.
- Contribute to irrigation water, farming water chestnuts by local villagers.



Pain areas of Sasur Khaderi River

- ~ 220 Km Long river however discontinuous at several locations and becoming extinct.
- Connecting to several water bodies including 5 ponds proposed by the Irrigation department.
- Lack of detailed hydrological, hydrogeological and other engineering data required for development of integrated watershed and river management plan and aquifer management plan.
- Encroachment in riverbed, flood plain in several locations throughout on the river stretch.
- Sewage and solid waste dumping in ponds and river stretches.



Farsi Akhnei Jheel

Project Components:

Component A: For Survey, Investigation & DPR Preparation:

DTM: Procurement of 5 m DTM

Drone Survey for the River & Ponds

- Drone data acquisition (Length of River = 220 Km)
- Ground Control Points (Each in 1 Km)
- Drone data processing DTM (For 220 Km Length & 0.5 Km on either side)

GIS Mapping (3D)

- Feature extraction and capturing all layers
- Encroachment mapping
- Cadastral mapping (river bed demarcation)

Geophysical Investigation

- Data from NGRI will be used.

Groundwater Mapping

- Consolidation of data
- Aquifer Mapping on 1:10000 scale

Data Assessment and Representation

- Ground water assessment
- Categorization of administrative units
- Aquifer response modelling
- Aquifer Management Strategies
- Preparation of groundwater development plan
- Development of Aquifer Management Plan (river Rejuvenation)

Hydrological Studies

- Data collection and Consolidation of Data
- Hydrological studies and modelling
- Preparation of integrated River rejuvenation plan

Detailed Project Report

Monitoring and Evaluation

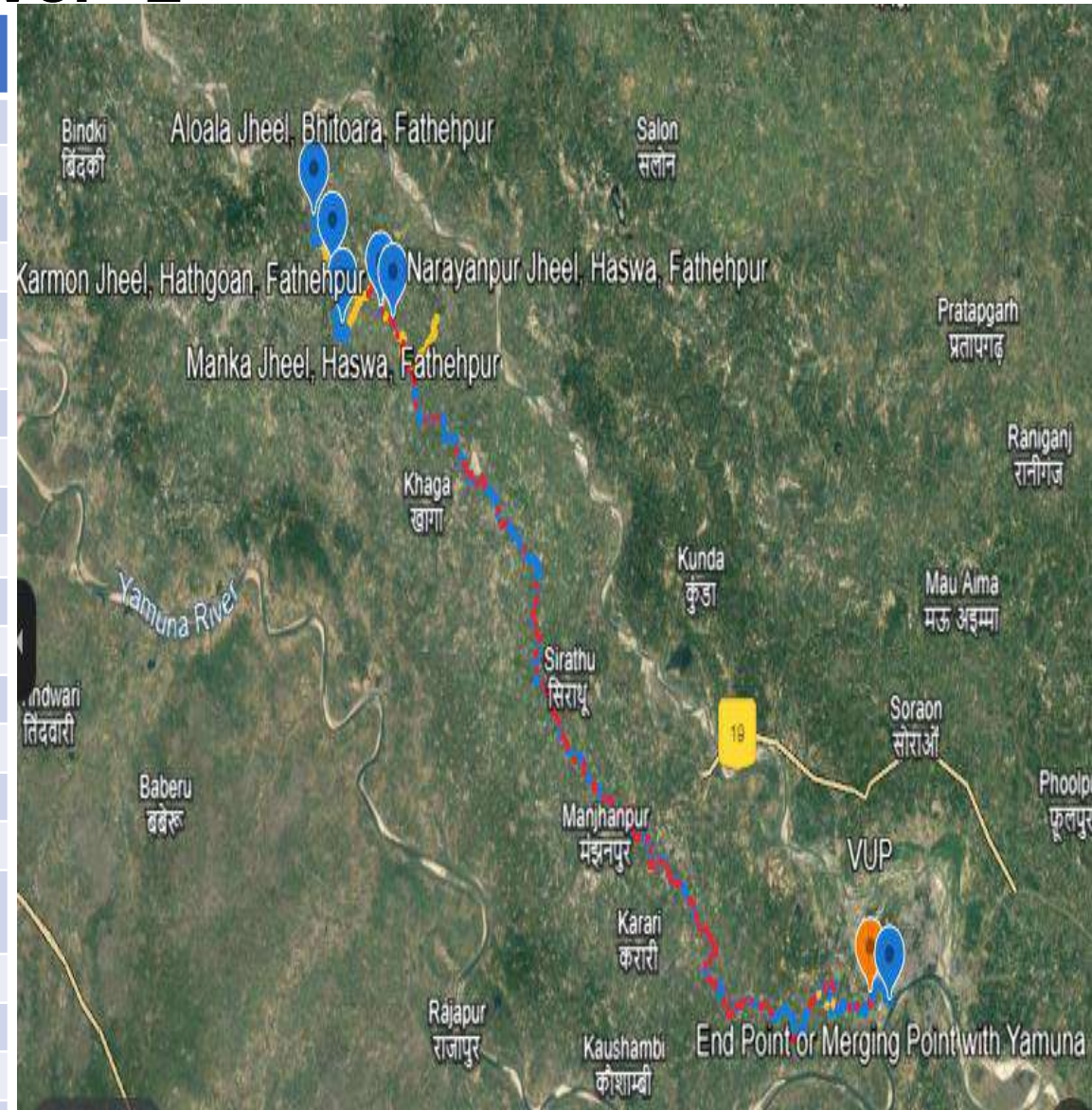
- Impact assessment of Pond rejuvenated after completion of Component B

Component B: Pilot project at Farsi Akhnei Jheel

- Deepening and de-silting of Farsi Akheni Jheel connected to the river system.
- Management plan for waterbody conservation
- Community participation and livelihood development for local population.
- State government through irrigation department to undertake “pond/ bandhe” project with the aim to ensure water availability and e flow to river throughout year.

Salient Feature of Sasur Khaderi River- 1

Name of River	:	Sasur Khaderi River No. 1		
Length of River	:	218.38 Kms		
Catchment Area	:	1201 Km ²		
Origin of River	:	Farsi Akhnei Jheel		
Elevation	:	135.0 m		
Tributary of	:	Yamuna River		
Merging at	:	Yamuna River near Prayagraj.		
Coordinates	:	Latitude: 25°24'18.39" N		
	:	Longitude: 81°48'31.55" E		
Elevation	:	94.0 m		
Structures Details	:			
Total	:	119		
VUP	:	2		
Weir	:	102		
Minor Bridges	:	9		
Culverts	:	5		
Other	:	Sasur Khaderi River SPS		
Ponds contributing water	:	1 Farsi Akhnei Jheel		
	:	2 Aloala Jheel		
	:	3 Karmon Jheel		
	:	4 Manka Jheel		
	:	5 Narayanpur Jheel		
Elevation Difference	:	41.0 m		
Slope	:	0.000188		

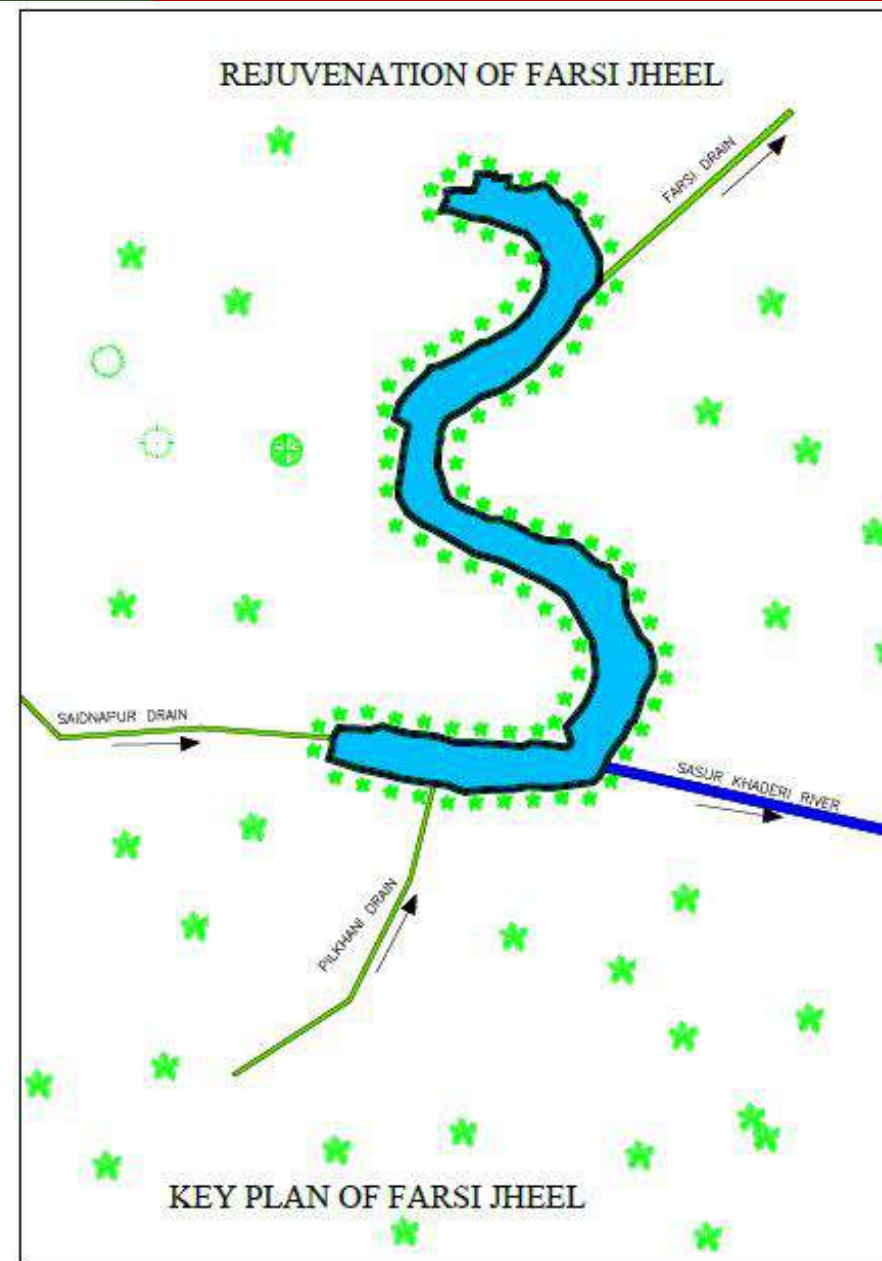


Pilot Project: Rejuvenation & conservation of Farsi Akhnai Jheel

Farsi Akhnai Lake: - Farsi Akhnai Lake is located in village Farsi, Mohddinpur, Ganpatpur, Lalpur, Kamalpur, Audhera, Pataria, Faridpur, Dharmadaspur and Pilikhini etc of bhिताura development block, Tehsil Fatehpur, district Fatehpur. The area of this lake is 1085968.09 square meters (108.60 Ha), at present the water holding capacity of the lake is 1476.92 ML. After the rejuvenation of the lake, the water holding capacity of the lake will be 3095.01 ML.

S. No.	Water Body Name	Location	Pond Area (Ha.)		Catchment Area (Ha.)	NMCG Catchment Area (Ha.)	Required Water Depth (m)	Excavation Depth (m)	Volume in Million Litre	
			Previous	Present					Present	After
1	Farsi Akhnei Jheel	Bhitoara, Fathehpur	133.91	108.6	1067.5	3814	4.92	3.56	1476.92	3095.01

S. No.	Water Body Name	Location	Catchment Area Villages	Inlet Drains	Outlet	Uses	Remarks
1	Farsi Akhnei Jheel	Bhitoara, Fathehpur	Farsi, Ganpatpur, Lalpur, Kamalpur, Faridpur, Dharamdaspur, Pilkhini, Pataria etc.	Sadnapur Nala & Pilkhini Nala	Through Sasur Khedri River-1 drop in Alola Jheel	Singara & Fish Farming, Local Irrigation, Fruits & Vegetable Farming	Encroachment, heavily silted



Scope of Work (Major Components)

Component A: For Survey, Investigation & DPR Preparation:

- **DTM:** Procurement of 5 m DTM
- **Drone Survey for the River & Ponds**
 - Drone data acquisition (Length of River = 220 Km)
 - Ground Control Points (Each in 1 Km)
 - Drone data processing DTM (For 220 Km Length & 0.5 Km on either side)
- **GIS Mapping**
 - Feature extraction and capturing all layers
 - Encroachment mapping
 - Cadastral mapping (river bed demarcation)
- **Geophysical Investigation**
 - Data from NGRI will be used.
- **Groundwater Mapping**
 - Consolidation of data
 - Aquifer Mapping on 1:10000 scale

Data Assessment and Representation

- Ground water assessment
- Categorization of administrative units
- Aquifer response modelling
- Aquifer Management Strategies
- Preparation of groundwater development plan

- Development of Aquifer Management Plan (river Rejuvenation)
- **Hydrological Studies**
 - Data collection and Consolidation of Data
 - Hydrological studies and modelling
 - Preparation of integrated River rejuvenation plan
- **Detailed Project Report**
- **Monitoring and Evaluation**
 - Impact assessment of Pond rejuvenated after completion of Component B

Component B: Pilot project at Farsi Akhnei Jheel

- Deepening and de silting of Farsi Akheni Jheel connected to the river system.
- Management plan for waterbody conservation
- Community participation and livelihood development for local population.
- State government through irrigation department to undertake “pond/ bandhe” project with the aim to ensure water availability and e flow to river throughout year

Estimated Cost:	Component A: Survey, investigation and DPR Cost – Rs. 338 Lakhs (Excluding GST) Component B: Pilot project at Farsi Akhnei Jheel Cost: 3733.90 lakhs O & M Cost for five years: 1,866.95 Lakhs (10% per year after one year) Total Cost : 5,938.85 lakhs
Central Government Share	Rs. 5,938.85 lakhs (100%)

Main Component Works taken for Pilot Project

- Earth work in strengthening bank of canal / Dam / Bundhies / Guide Bundh/ Embankment by hydraulic excavator (Poclain/JCB) including proper compaction in layers and dressing by mechanical equipment.
- Dressing of earth in canal banks, bunds and daula etc.
- Brick on edge soling with straight over burnt brick including preparation of sub grade to prescribed camber and including supply of bricks filling joints with local earth spearding sand over the entire surface.
- Supplying & Fixing of NP-2 R.C.C Hume pipe 900 mm dia in inlet i/c cost of all material, labour etc.
- Supplying & laying local sand in filling under brick soling i/c cost of all material, labour etc. for proper complition of work.
- Plantation work around the jheel on bund i/c cost of all material, labour etc. for proper complition of work.
- Turfing with doob grass including watering

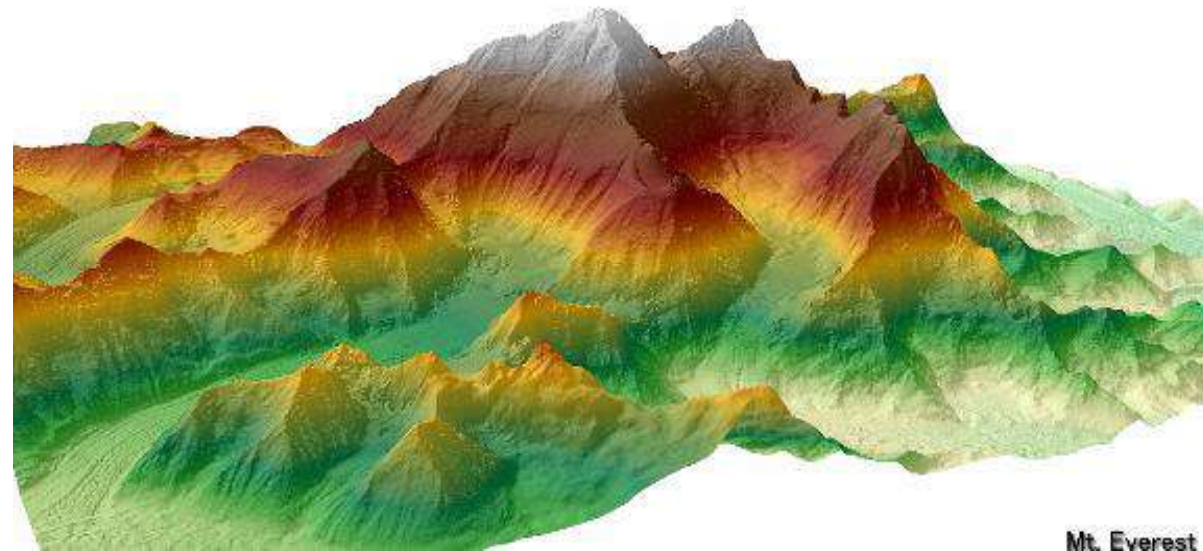
For Calculating required depth of pond for collecting catchment storm water, we analysis yearly daily rainfall data of Fatehpur District & calculate max. design rainfall for **100- yr return period**.

Village/ Town Name - Fatehpur			
S.No .	Year	Date	Daily Max. Railfall in mm
1	2012	25.07.2012	91.8
2	2013	19.06.2013	286.2
3	2014	22.07.2014	38.2
4	2015	12.07.2015	14.8
5	2016	17.08.2016	96.8
6	2017	10.08.2017	96.8
7	2018	25.08.2018	94.8
8	2019	28.09.2019	128.4
9	2020	22.07.2020	81.8
10	2021	24.09.2021	140

For Fatehpur				
S. No.	Return Period in Yrs.	Design Rainfall in mm		
		Grumble	Log Pearson III	Excel- Log Pearson III
1	2	97	98	
2	5	184	166	
3	10	242	205	
4	25	315	244	165.00
5	50	369	268	306.10
6	100	423	286	560.00
7	200	477	302	

Approach & Methodology

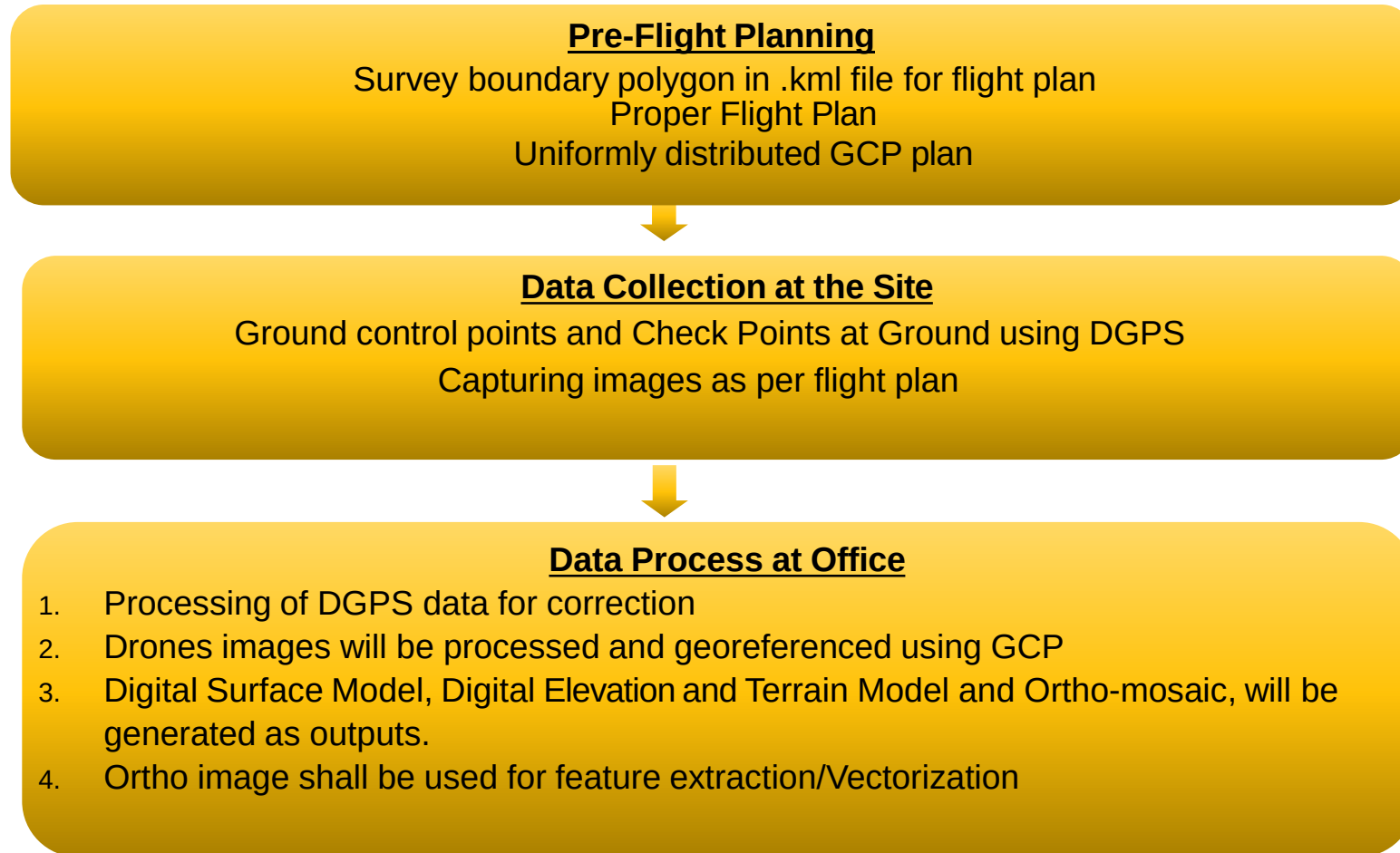
- Digital Terrain Model using stereo pair of Satellite image data
- Digital photogrammetry is an advance tool used to build models of the earth's surface that can produce high resolution DTMs by image matching automatically.
- The dimension of internal DTMs are measured in pixels and are related to the camera metrics quality, photo scale, flight height, Ground Sample Distance (GSD), surface morphology, vegetation, shadows, and atmospheric conditions.
- Digital photogrammetry can produce a 3D height point with high spatial resolution, and the image correlation method is applied to DTM extraction automatically.
- DTM (5 meter accuracy) is proposed for 1201 sq km covering the catchment of River Sasur Khaderi



Mt. Everest

Approach & Methodology

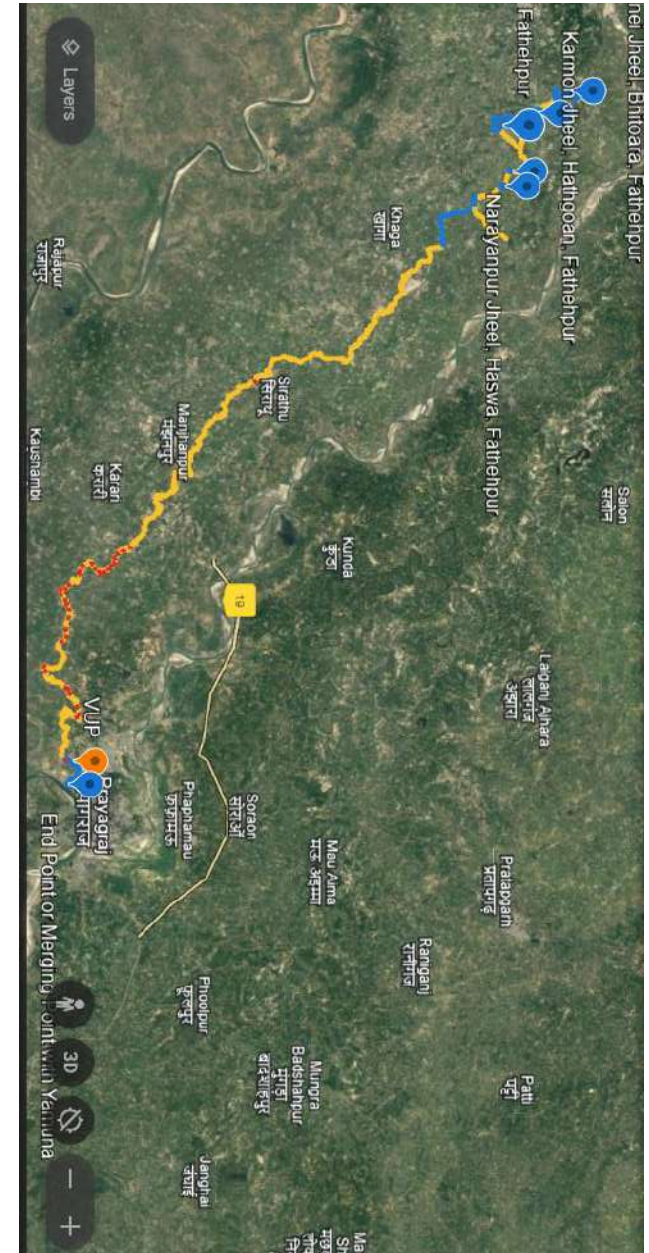
A. Drone-based Photogrammetric Survey – very high resolution for flood plain and river bed



Work Flow Diagram

Ground Control Points

- Ground control survey involves the precise measurement and establishment of ground reference points with known coordinates on the Earth's surface. These reference points, known as Ground Control Points (GCPs), serve as anchor points to accurately geo reference spatial data collected from remote sensing technologies like drones, satellites, or aerial photography.
 - Accuracy Enhancement
 - Geo-referencing
 - Quality Assurance
- **Process of Ground Control Survey:**
 - Selection of GCPs
 - Surveying GCPs using high-precision surveying instruments like DGPS
 - Field Verification
 - Geo-referencing of aerial photos



GCP Plan

Mission Planning

Mission planning is a meticulous process that involves defining objectives, selecting flight parameters, and ensuring safe and efficient data collection.

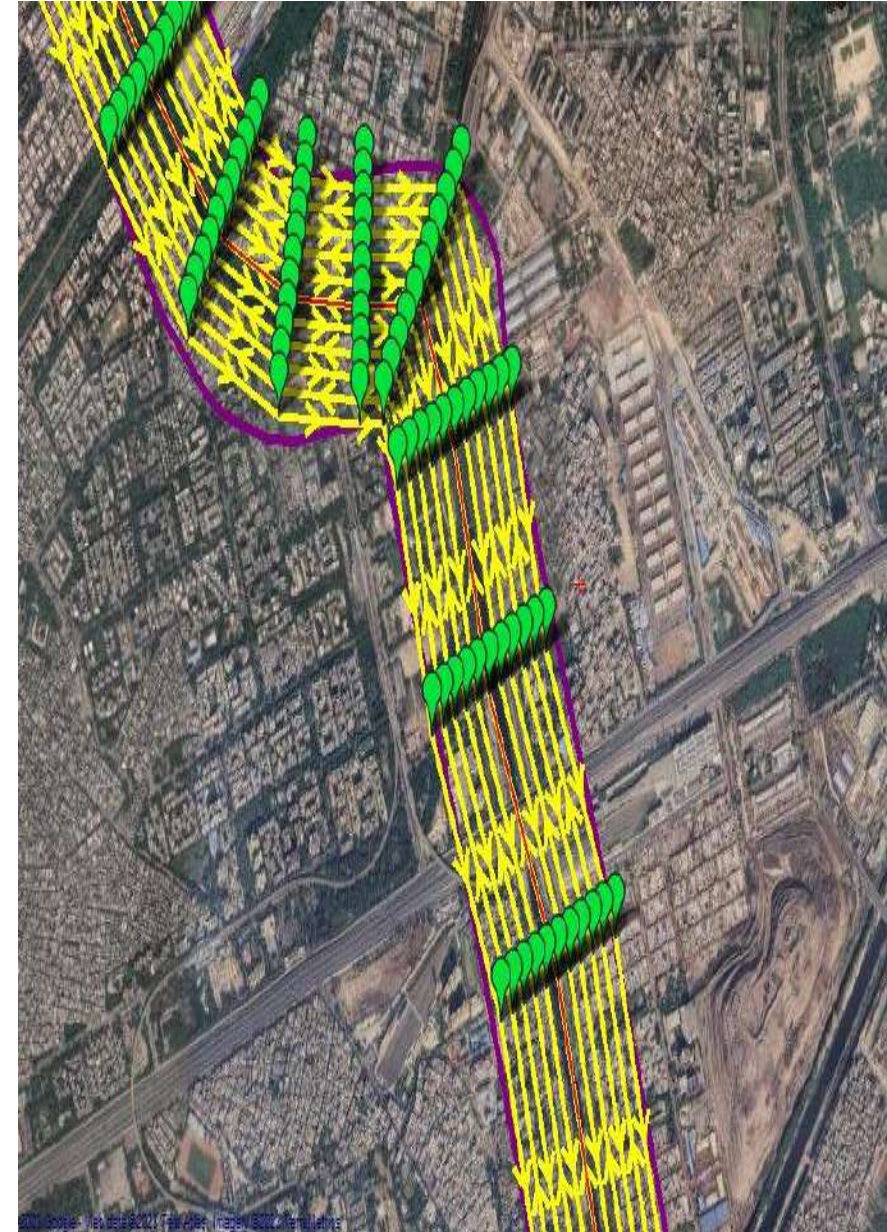
Area Analysis: Understand its topography, obstacles, and potential hazards. Identifying features like power lines, buildings, and trees helps you design a safe and effective flight path.

Flight Parameters: Key parameters include flight altitude, speed, and camera settings. Higher altitudes can cover larger areas but may reduce ground resolution, so striking the right balance is essential.

Image Overlap: Optimal image overlap is crucial for generating accurate 3D models and maps. Flight Path Design:

Safety First: Prioritize safety at all stages of mission planning. Incorporate emergency protocols and precautions for unexpected situations.

Battery Management: Plan for battery swaps or recharges if necessary and consider battery life reductions due to weather conditions.

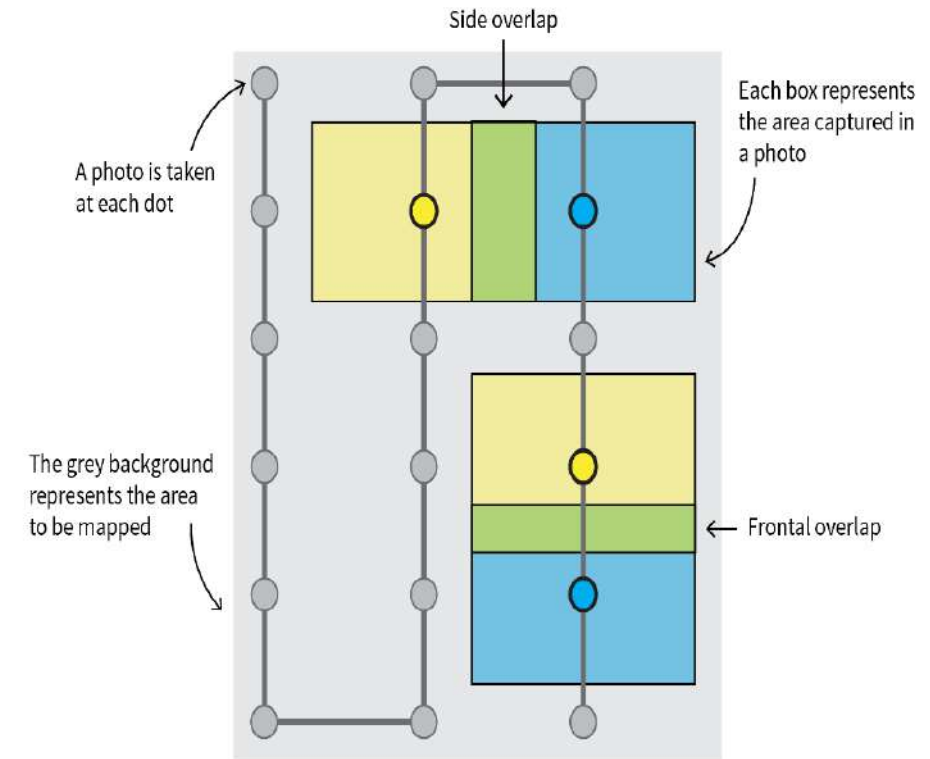
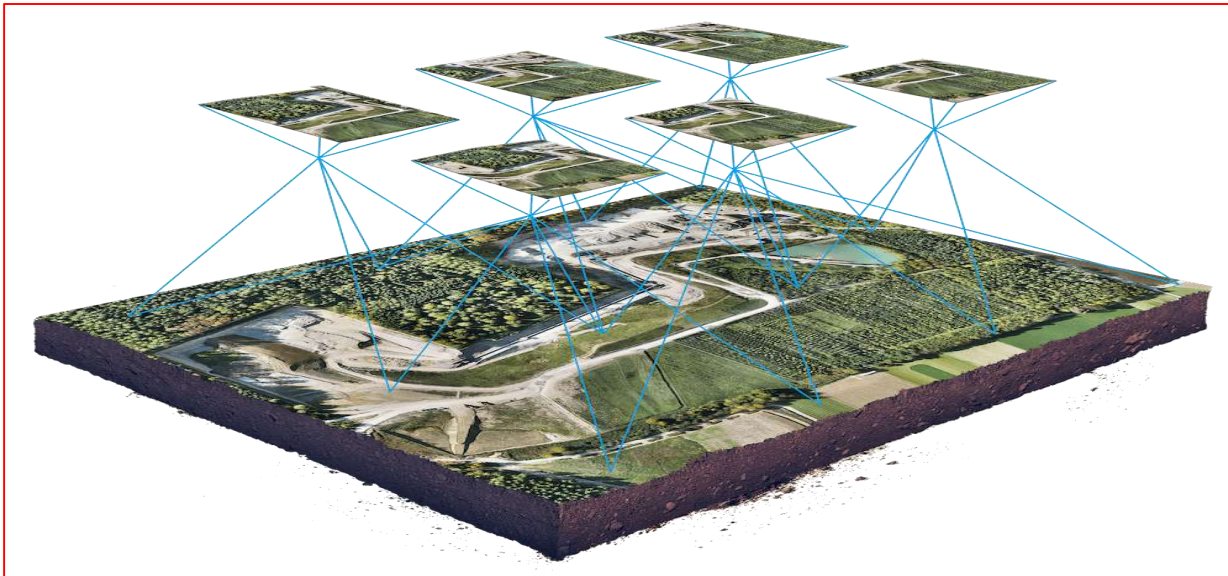


Drone Data acquisition

Data Capturing: Involves capturing a multitude of high-resolution images from various angles.

Overlap and Grid Patterns: Ensure that the images captured overlap significantly, facilitating accurate reconstruction of the landscape.

Vertical and Oblique Imagery: Drones capture both vertical (straight down) and oblique (angled) images. Vertical images aid in creating ortho-mosaic maps, while oblique images capture terrain details from different perspectives, enhancing the 3D model's realism.



Drone Data Processing

Image Pre-processing: Include correcting lens distortion, removing vegetating, adjusting exposure, and color balancing to enhance the overall image quality.

Image Orientation: accurate position and orientation of the camera for each image, known as exterior orientation parameters (EOP).

Feature Extraction: As reference points for triangulation. ie Point Identification:

Matching and Bundle Adjustment Bundle adjustment techniques are then used to optimize the camera parameters and 3D positions of the tie points, considering the constraints of multiple overlapping images.

Triangulation: Once the camera parameters and tie points' 3D positions are refined, aerial triangulation is performed.

Ground Control Points (GCP) Refinement: In order to improve the accuracy of the triangulation results,

Dense Point Cloud Generation: involves estimating the 3D coordinates of a large number of points on the ground surface

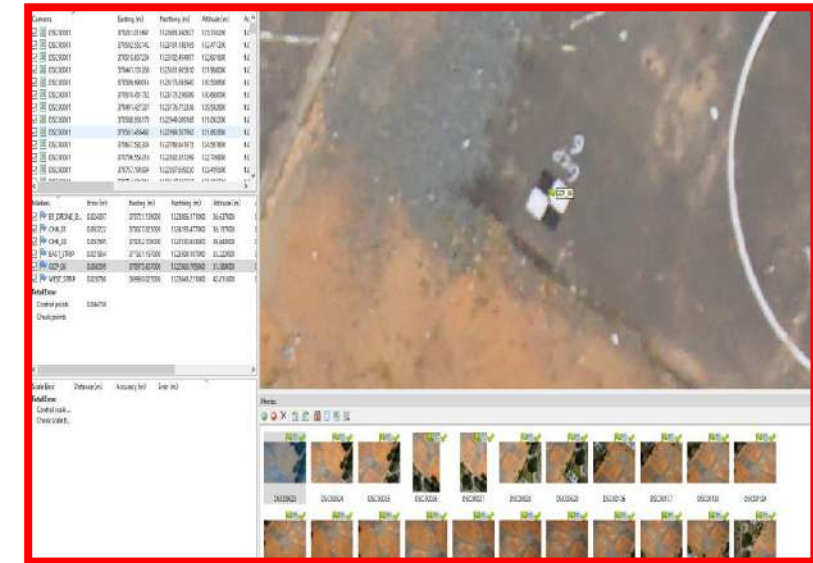
Digital Surface Model (DSM) Refinement: Refine the DSM by interpolating elevation values and improving the resolution,.

Digital Terrain Model (DTM) Generation: Use the filtered point cloud to generate a Digital Terrain Model (DTM)

Ortho-mosaic Color Correction: Perform color correction on the ortho-mosaic to improve visual representation and consistency.

Ortho-mosaic Export: Export the ortho-mosaic in the desired format (e.g., GeoTIFF) with embedded geospatial information.

Quality Assessment: Conduct a thorough quality assessment of all the final products, including the DSM, DTM, and ortho-mosaic.



GCP Refinement

OUTPUTS:

Orthomosaic Maps: An orthomosaic map is a high-resolution, georeferenced image that removes distortions and perspective effects, providing a true-to-scale representation of the surveyed area.

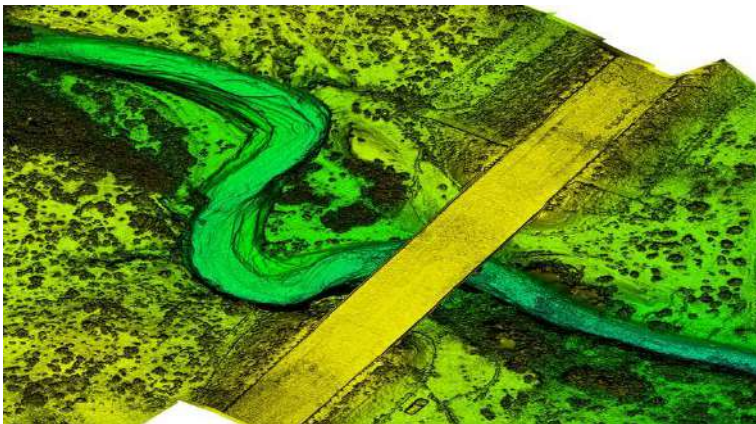
Digital Elevation Models (DEMs): DEMs are 3D representations of the Earth's surface, showing elevation variations across the surveyed area. DEMs are used for terrain analysis, slope calculations, and flood modeling.

Digital Surface Models (DSMs): DSMs include not only the ground surface but also any above-ground features like buildings, trees, and structures. DSMs are useful for urban planning, line-of-sight analysis, and visualizations.

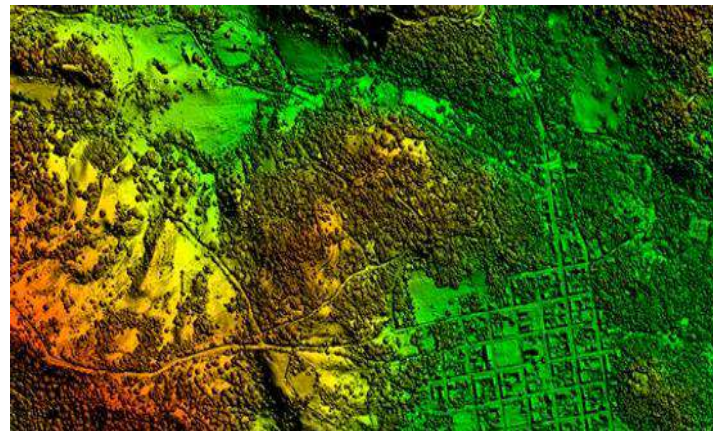
TOPOMAPS: including terrain, elevation, natural and human-made features, water bodies, vegetation, and more.



OTHO



DEM



DSM



TOPO Map

GIS Mapping

GIS Mapping (2D) - Feature Extraction, Encroachment Mapping, Cadastral Mapping

1.Feature Extraction: a. **Identify Relevant Features:**

2. Feature Classification: land cover, water bodies, buildings, roads

3.Encroachment Mapping: legal and environmental perspectives.

4.Data Collection: Utilize available datasets, cadastral maps, satellite imagery, and ground surveys to identify and document encroachments along the river bed and its vicinity.

5.Mapping and Analysis: Type, extent, and potential impact on the environment.

6.Legal Boundaries: Accurate identification of legal boundaries and encroachment limits.

7.Cadastral Mapping (River Bed Demarcation):

8.Boundary Survey: Conduct a detailed boundary survey of the river bed area to establish accurate cadastral boundaries.

9.Parcel Identification: Identify and mark cadastral parcels within the river bed, considering legal, environmental, and land-use parameters.

10.Data Integration: Integrate cadastral mapping data with other GIS layers, ensuring seamless representation of cadastral information within the overall GIS mapping system.

11.Quality Control and Assurance: a. **Accuracy Assessment:**

12.Validation: Verify the accuracy of GIS mapping results against ground truth data.

13.Documentation and Reporting: a. **Prepare Detailed Reports:**

14. Recommendations: Recommendations for mitigating encroachments and improving cadastral mapping accuracy.

Groundwater Mapping and Assessment

1. **Groundwater Mapping:** a. **Data Compilation:** Existing groundwater data, including well logs, hydrogeological reports, and other relevant sources.
2. **GIS Integration:**
3. **Aquifer Identification:** Identify and delineate aquifer boundaries
4. **Hydrogeological Zoning:** Based on groundwater availability, quality, and vulnerability.
5. **Groundwater Assessment:** a. **Aquifer Parameters:** Hydraulic conductivity, transmissivity, storativity, and recharge rates.
6. **Groundwater Quality :** pH, total dissolved solids (TDS), and major ion concentrations.
7. **Water Level Monitoring:** Establish a network for continuous monitoring of groundwater levels to understand seasonal fluctuations and trends.
8. **Sustainability Analysis:** Extraction rates, recharge rates, and environmental impacts.
9. **Risk Assessment:** To identify areas prone to groundwater depletion or contamination.
10. **Aquifer Response Modeling:** a. **Model Development:** Develop numerical models to simulate aquifer response to various stressors, such as pumping, recharge, and land-use changes.
11. **Scenario Analysis:** To understand how different management strategies impact aquifer response.
12. c. **Predictive Modeling:** To forecast future groundwater levels and quality under different scenarios.
13. **Aquifer Management Strategies:** a. **Regulatory Measures:** for sustainable groundwater use, including pumping restrictions and well-spacing guidelines.
14. **Recharge Enhancement:** For enhancing aquifer recharge through natural and artificial means.
15. **Community Engagement:** Involve local communities in groundwater management decisions and encourage responsible water use practices.
16. **Preparation of Groundwater Development Plan:** a. **Infrastructure Recommendations:** for groundwater infrastructure development, including well construction and maintenance.
17. **Development of Aquifer Management Plan (River Rejuvenation):** a. **Integrated Approach:** Integrate aquifer management with river rejuvenation plans to address both surface and subsurface water interactions.
18. **Ecosystem Restoration:** For restoring river ecosystems, considering groundwater-surface water interactions.
19. **Monitoring and Adaptive Management:** To track the effectiveness of the aquifer management plan and make adjustments as needed.
20. **Documentation and Reporting:** a. **Technical Reports:**

Hydrological Studies

1.Data Collection and Consolidation of Data: Meteorological stations, river gauge stations, satellite imagery, and any available historical hydrological data.

2.Field Surveys: To gather ground-truth data, including river discharge measurements, water quality samples, and bathymetric surveys of ponds.

3.Data Validation: for accuracy and consistency.

4.Hydrological Studies and Modeling: a. **Hydrological Modeling** :to simulate river flow, precipitation runoff, and groundwater interactions. b.

Flood Risk Assessment: Modeling different scenarios, including extreme weather events and potential changes in land

5. Water Balance Analysis: to understand the overall water availability and consumption in the study area.

6. Climate Change Impact: Incorporate climate change scenarios into hydrological models to assess potential impacts on the hydrological cycle.

1.Preparation of Integrated River Rejuvenation Plan: a. **Identification of Intervention Areas:** for ecological restoration and sustainable water management.

2. Erosion Control Measures: To control soil erosion along riverbanks.

3. Riparian Zone Protection: Develop strategies for protecting and enhancing riparian zones to improve water quality and biodiversity.

4. Water Quality Improvement: Recommend measures to address pollution sources and enhance water quality.

5.Detailed Project Report: a. **Engineering Designs:** Develop detailed engineering designs for proposed interventions, such as check dams, riverbank stabilization, and pond rejuvenation.

6. b. Cost Estimates: Provide detailed cost estimates for implementing the proposed projects.

7.c. Timeline and Phasing: Outline a project timeline and phasing strategy for the implementation of proposed interventions.

8. Monitoring and Evaluation: a. **Monitoring Framework:**

1. Develop a comprehensive monitoring framework to track the progress of implemented projects.

2. Data Collection and Analysis: Continuously collect data on key indicators, such as river discharge, water quality, and ecological health.

3. Adaptive Management: Implement an adaptive management approach, making adjustments to interventions based on real-time monitoring data.

4. Stakeholder Engagement: Involve local communities and stakeholders in the monitoring process to gather valuable feedback.

9.Impact Assessment of Pond Rejuvenated: a. **Baseline Assessment:** Establish baseline conditions for the ponds, including water quality, biodiversity, and overall ecological health.

1. Implementation Impact: Assess the impact of pond rejuvenation projects on water storage, groundwater recharge, and local ecosystems.

2. Socioeconomic Impact: Evaluate the socioeconomic impact of pond rejuvenation on local communities, including changes in livelihoods and water availability.

3. Documentation and Reporting: Prepare a comprehensive impact assessment report, documenting findings and recommendations for